

Optional Exercises II

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1 Graph Theory

Problem 1: Prove that every complete graph with $n > 2$ vertices contains a cycle of length n .

Problem 2: Let G be a (not necessarily connected) graph where the length of the longest path is 2. Prove that G is bipartite.

Problem 3: How many length 4 paths exist on a complete graph with n vertices?

Problem 4: What is the largest number of edges that a graph with n vertices can have and still be 2-colorable?